Intelligent Robots

LAB1

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OBJECTIVES

01 Lab introduction

02 Install Ubuntu 18.04

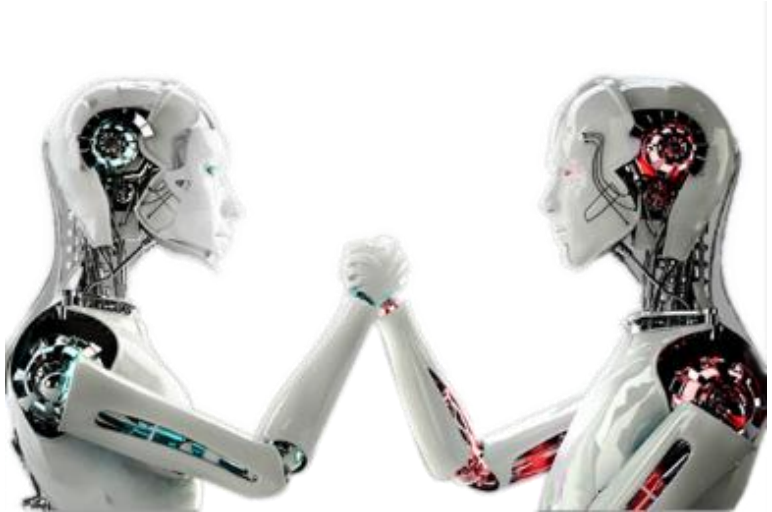
03 Basic operation of Linux system

04 The basics of C++ & python

05 Installing ROS

PART ONE

Lab Introduction



The development platform and tools involved

Ubuntu

Ubuntu is the most popular Linux distribution in the world. It is an open source software operating system that runs from the desktop, to the cloud, to all your internet connected things.

The Robot Operating System (ROS)

The Robot Operating System (ROS) is a flexible framework for writing robot software.

http://www.ros.org

Turtlebot

TurtleBot is a low-cost, personal robot kit with open-source software.
With TurtleBot, you'll be able to build a robot that can drive around your house, see in 3D, and have enough horsepower to create exciting applications. http://wiki.ros.org/Robots/TurtleBot

MATLAB Robotics System Toolbox

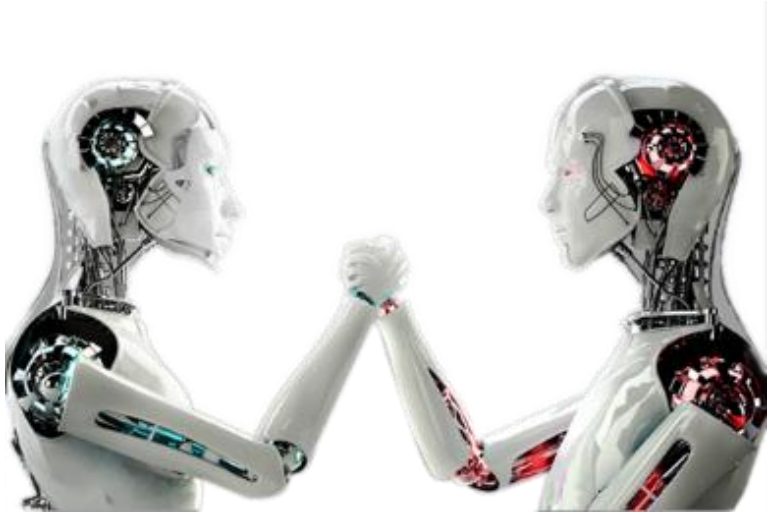
Robotics System Toolbox™ provides tools and algorithms for designing, simulating, and testing manipulators, mobile robots, and humanoid robots.
https://ww2.mathworks.cn/help/robotics/index.html?searchHighlight=robotics&s_tid=doc_srchtitle

Gazebo

Gazebo is a 3D dynamic simulator with the ability to accurately and efficiently simulate populations of robots in complex indoor and outdoor environments . http://gazebosim.org

PART TWO

Install Ubuntu



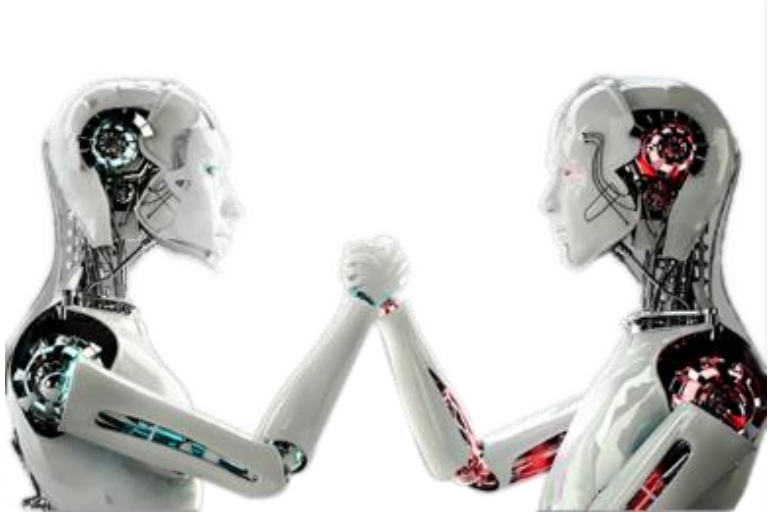
Install Ubuntu

How to Dual Boot Ubuntu 18.04 LTS with Windows 10:

Ubuntu 18.04 LTS Desktop Installation Guide:

PART THREE

Basic operation of Linux system



Basic operation of Linux system

●Basic linux commands

No.	Command	Description	No.	Command	Description
1	pwd	Display the pathname for the current directory.	9	touch	Create an empty file with the specified name.
2	ls	List directory contents.	10	clear	Clear a command line screen/window for a fresh start.
3	cd	Change to directory.	11	find	Finds the file in the specified directory
4	cp	Copy files and directories.	12	sudo	A command requires superuser privileges
5	mv	Rename or move file(s) or directories.	13	chmod	Change a file’s permissions.
6	rm	Remove (delete) file(s) and/or directories	14	ll	List the detail directory contents
7	mkdir	Create a new directory.	15	ifconfig	view or configure a network interface
8	rmdir	Delete empty directories.	16	history	Lists previously used commands

Basic operation of Linux system

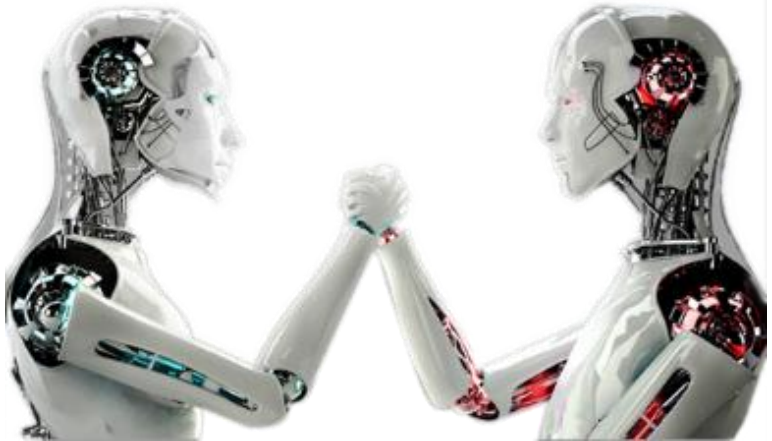
- **Basic linux commands**

```
jyh@jyh:~$ pwd
/home/jyh
jyh@jyh:~$ ls
Desktop    Downloads         Music     Public     Videos
Documents  examples.desktop  Pictures  Templates
jyh@jyh:~$ mkdir testdir
jyh@jyh:~$ ls
Desktop    Downloads         Music     Public     testdir
Documents  examples.desktop  Pictures  Templates  Videos
jyh@jyh:~$ cd testdir/
jyh@jyh:~/testdir$ ls
jyh@jyh:~/testdir$ touch test
jyh@jyh:~/testdir$ ls
test
jyh@jyh:~/testdir$ mv test ../mvtest
jyh@jyh:~/testdir$ cd ..
jyh@jyh:~$ ls
Desktop    Downloads         Music   Pictures  Templates  Videos
Documents  examples.desktop  mvtest  Public    testdir
jyh@jyh:~$ rm mvtest
jyh@jyh:~$ ls
Desktop    Downloads         Music     Public     testdir
Documents  examples.desktop  Pictures  Templates  Videos
jyh@jyh:~$ cd testdir/
jyh@jyh:~/testdir$ ls
jyh@jyh:~/testdir$ touch test
jyh@jyh:~/testdir$ ls
test
jyh@jyh:~/testdir$ ll
total 8
drwxr-xr-x  2 jyh jyh 4096 2月  16 21:35 ./
drwxr-xr-x 16 jyh jyh 4096 2月  16 21:34 ../
-rw-r--r--  1 jyh jyh    0 2月  16 21:35 test
jyh@jyh:~/testdir$ chmod +x test
jyh@jyh:~/testdir$ ll
total 8
drwxr-xr-x  2 jyh jyh 4096 2月  16 21:35 ./
drwxr-xr-x 16 jyh jyh 4096 2月  16 21:34 ../
-rwxr-xr-x  1 jyh jyh    0 2月  16 21:35 test*
```

Refer to <Basic operation of Linux system.mp4>

PART FOUR

Installing ROS



What is ROS and why do we learn ROS?

The Robot Operating System (ROS) is a flexible framework for writing robot software. It is a collection of tools, libraries that aim to simplify the task of creating complex and robust robot behavior across a wide variety of robotic platforms.

History of ROS

2007
诞生于斯坦福
STAIR项目
Morgan Quigley

Box Turtle

2010
ROS 1.0 发布

ROSCon 2012

2012
第一届ROScon

2014
ROS Indigo发布

2017
ROS 2.0 Ardent发布

2008
Willow Garage接手

2011
TurtleBot发布

2013
OSRF接管

Open Source Robotics Foundation

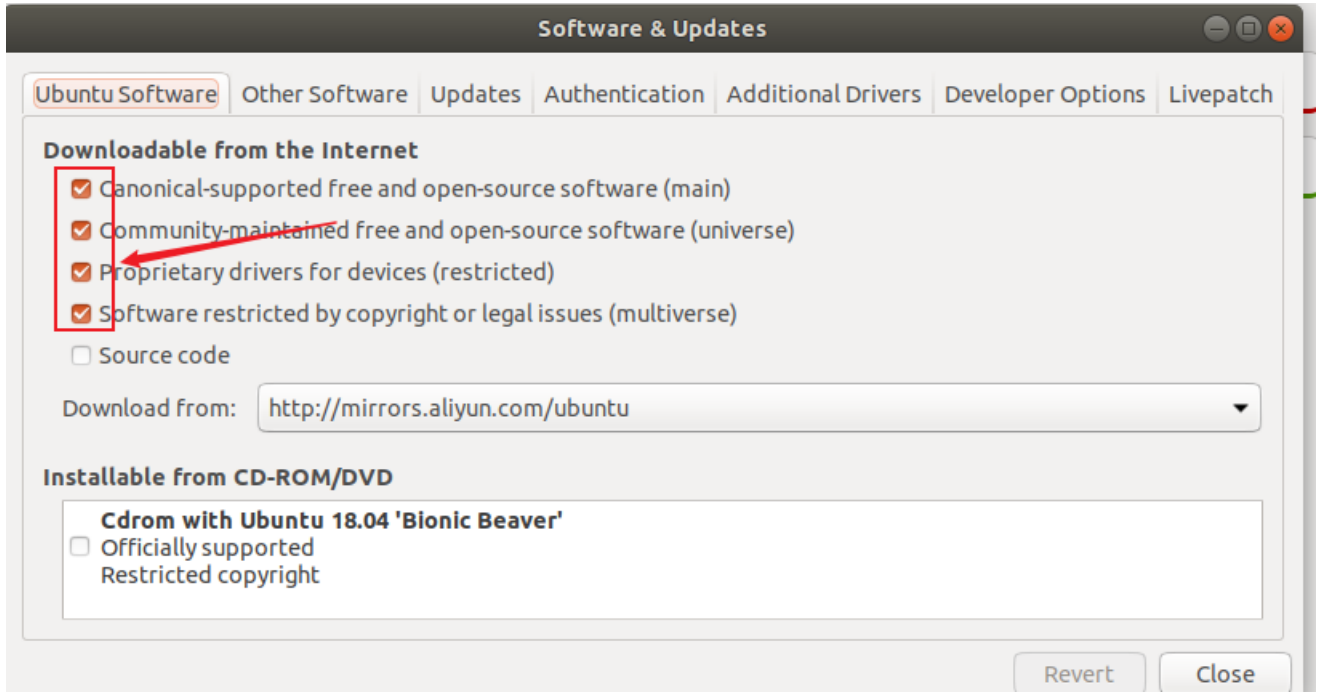
2016
ROS Kinetic发布

2018
ROS Melodic发布

ROS Installation

Refer to http://wiki.ros.org/melodic/Installation/Ubuntu for ROS installation

Step 1.Verify your Ubuntu repositories



Step 2. Setup your sources.list

$sudo sh -c '. /etc/lsb-release && echo "deb http://mirrors.ustc.edu.cn/ros/ubuntu/ `lsb_release -cs` main" > /etc/apt/sources.list.d/ros-latest.list'

ROS Installation

Step 3.Set up your keys

$sudo apt-key adv --keyserver 'hkp://keyserver.ubuntu.com:80' --recv-key C1CF6E31E6BADE8868B172B4F42ED6FBAB17C654

Step 4. Installation

$sudo apt-get update

```
jyh@jyh:~$ sudo apt-get update
[sudo] password for jyh:
Hit:1 http://mirrors.aliyun.com/ubuntu bionic InRelease
Get:2 http://mirrors.aliyun.com/ubuntu bionic-updates InRelease [88.7 kB]
Get:3 http://mirrors.aliyun.com/ubuntu bionic-backports InRelease [74.6 kB]
Get:4 http://mirrors.aliyun.com/ubuntu bionic-security InRelease [88.7 kB]
Get:5 http://mirrors.aliyun.com/ubuntu bionic-updates/main amd64 Packages [869 kB]
Get:6 http://mirrors.aliyun.com/ubuntu bionic-updates/main i386 Packages [648 kB]
Get:7 http://mirrors.aliyun.com/ubuntu bionic-updates/main Translation-en [302 kB]
Get:8 http://mirrors.aliyun.com/ubuntu bionic-updates/main amd64 DEP-11 Metadata [307 kB]
Get:9 http://mirrors.aliyun.com/ubuntu bionic-updates/main DEP-11 48x48 Icons [73.8 kB]
Get:10 http://mirrors.aliyun.com/ubuntu bionic-updates/main DEP-11 64x64 Icons [140 kB]
Get:11 http://mirrors.aliyun.com/ubuntu bionic-updates/universe amd64 Packages [1,055 kB]
Get:12 http://mirrors.aliyun.com/ubuntu bionic-updates/universe i386 Packages [1,012 kB]
Get:13 http://mirrors.aliyun.com/ubuntu bionic-updates/universe Translation-en [327 kB]
Get:14 http://mirrors.aliyun.com/ubuntu bionic-updates/universe amd64 DEP-11 Metadata [264 kB]
Get:15 http://mirrors.aliyun.com/ubuntu bionic-updates/universe DEP-11 48x48 Icons [187 kB]
Get:16 http://mirrors.aliyun.com/ubuntu bionic-updates/universe DEP-11 64x64 Icons [432 kB]
Get:17 http://mirrors.aliyun.com/ubuntu bionic-updates/multiverse amd64 DEP-11 Metadata [2,464 B]
Get:18 http://mirrors.aliyun.com/ubuntu bionic-backports/universe amd64 DEP-11 Metadata [8,284 B]
Get:19 http://mirrors.aliyun.com/ubuntu bionic-security/main amd64 Packages [650 kB]
Get:20 http://mirrors.aliyun.com/ubuntu bionic-security/main i386 Packages [442 kB]
Get:21 http://mirrors.aliyun.com/ubuntu bionic-security/main amd64 DEP-11 Metadata [38.5 kB]
Get:22 http://mirrors.aliyun.com/ubuntu bionic-security/main DEP-11 64x64 Icons [41.5 kB]
Get:23 http://mirrors.aliyun.com/ubuntu bionic-security/universe amd64 Packages [648 kB]
Get:24 http://mirrors.aliyun.com/ubuntu bionic-security/universe i386 Packages [617 kB]
Get:25 http://mirrors.aliyun.com/ubuntu bionic-security/universe amd64 DEP-11 Metadata [42.0 kB]
Get:26 http://mirrors.aliyun.com/ubuntu bionic-security/universe DEP-11 48x48 Icons [16.4 kB]
Get:27 http://mirrors.aliyun.com/ubuntu bionic-security/universe DEP-11 64x64 Icons [111 kB]
Get:28 http://mirrors.aliyun.com/ubuntu bionic-security/multiverse amd64 DEP-11 Metadata [2,460 B]
Hit:29 http://mirrors.ustc.edu.cn/ros/ubuntu bionic InRelease
Fetched 8,488 kB in 10s (830 kB/s)
Reading package lists... Done
```

$sudo apt install ros-melodic-desktop-full

```
jyh@jyh:~$ sudo apt-get install ros-melodic-desktop-full
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following packages were automatically installed and are no longer required:
  efibootmgr gir1.2-geocodeglib-1.0 libfwup1 ubuntu-web-launchers
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  autoconf automake autopoint autotools-dev binfmt-support blt build-essential bzip2-doc cmake cmake-data curl cython debhelper
  default-libmysqlclient-dev dh-autoreconf dh-python dh-strip-nondeterminism docutils-common docutils-doc dpkg-dev fakeroot
  fltk1.3-doc fluid fonts-lato fonts-lyx freeglut3 freeglut3-dev g++ g++-7 gazebo9 gazebo9-common gazebo9-plugin-base gcc
  gcc-4.8-base gcc-7 gdal-data gir1.2-gtk-2.0 gir1.2-harfbuzz-0.0 google-mock googletest graphviz hddtemp hdf5-helpers
  ibverbs-providers icu-devtools javascript-common libaec-dev libaec0 libalgorithm-diff-perl libalgorithm-diff-xs-perl
  libalgorithm-merge-perl libann0 libapr1 libapr1-dev libaprutil1 libaprutil1-dev libarchive-cpio-perl libarmadillo-dev
  libarmadillo8 libarpack2 libarpack2-dev libasan0 libasan4 libassimp-dev libassimp4 libassuan-dev libatk1.0-dev libatomic1
  libavcodec-dev libavdevice-dev libavdevice57 libavfilter-dev libavformat-dev libavresample-dev libavutil-dev libblas-dev
  libblas3 libboost-all-dev libboost-atomic-dev libboost-atomic1.65-dev libboost-atomic1.65.1 libboost-chrono-dev
  libboost-chrono1.65-dev libboost-chrono1.65.1 libboost-container-dev libboost-container1.65-dev libboost-container1.65.1
  libboost-context-dev libboost-context1.65-dev libboost-context1.65.1 libboost-coroutine-dev libboost-coroutine1.65-dev
  libboost-coroutine1.65.1 libboost-date-time-dev libboost-date-time1.65-dev libboost-dev libboost-exception-dev
  libboost-exception1.65-dev libboost-fiber-dev libboost-fiber1.65-dev libboost-fiber1.65.1 libboost-filesystem-dev
```

```
  ros-melodic-transmission-interface ros-melodic-turtle-actionlib ros-melodic-turtle-tf ros-melodic-turtle-tf2
  ros-melodic-turtlesim ros-melodic-urdf ros-melodic-urdf-parser-plugin ros-melodic-urdf-sim-tutorial ros-melodic-urdf-tutorial
  ros-melodic-urdfdom-py ros-melodic-vision-opencv ros-melodic-visualization-marker-tutorials ros-melodic-visualization-msgs
  ros-melodic-visualization-tutorials ros-melodic-viz ros-melodic-webkit-dependency ros-melodic-xacro ros-melodic-xmlrpcpp ruby
  ruby-did-you-mean ruby-minitest ruby-net-telnet ruby-power-assert ruby-test-unit ruby2.5 rubygems-integration sbcl
  sdformat-sdf sgml-base sip-dev tango-icon-theme tcl tcl-dev tcl-vtk6 tcl8.6 tcl8.6-dev tk tk-dev tk8.6 tk8.6-blt2.5 tk8.6-dev
  ttf-bitstream-vera ttf-dejavu-core unixodbc-dev uuid-dev vtk6 x11proto-composite-dev x11proto-core-dev x11proto-damage-dev
  x11proto-dev x11proto-fixes-dev x11proto-input-dev x11proto-randr-dev x11proto-scrnsaver-dev x11proto-xext-dev
  x11proto-xf86vidmode-dev x11proto-xinerama-dev xml-core xorg-sgml-doctools xtrans-dev zlib1g-dev
0 upgraded, 1021 newly installed, 0 to remove and 7 not upgraded.
Need to get 496 MB of archives.
After this operation, 2,243 MB of additional disk space will be used.
Do you want to continue? [Y/n] Y
Get:1 http://mirrors.aliyun.com/ubuntu bionic/main amd64 fonts-lato all 2.0-2 [2,698 kB]
Get:2 http://mirrors.aliyun.com/ubuntu bionic/universe amd64 libtinyxml2.6.2v5 amd64 2.6.2-4 [29.7 kB]
Get:3 http://mirrors.aliyun.com/ubuntu bionic/universe amd64 libopenni0 amd64 1.5.4.0-14build1 [276 kB]
Get:4 http://mirrors.aliyun.com/ubuntu bionic/main amd64 libdouble-conversion1 amd64 2.0.1-4ubuntu1 [33.0 kB]
Get:5 http://mirrors.aliyun.com/ubuntu bionic-updates/main amd64 libqt5core5a amd64 5.9.5+dfsg-0ubuntu2.5 [2,036 kB]
Get:6 http://mirrors.aliyun.com/ubuntu bionic-updates/main amd64 libqt5dbus5 amd64 5.9.5+dfsg-0ubuntu2.5 [195 kB]
Get:7 http://mirrors.aliyun.com/ubuntu bionic-updates/main amd64 libqt5network5 amd64 5.9.5+dfsg-0ubuntu2.5 [635 kB]
Get:8 http://mirrors.aliyun.com/ubuntu bionic-updates/main amd64 libxcb-xinerama0 amd64 1.13-2~ubuntu18.04 [5,264 B]
Get:9 http://mirrors.aliyun.com/ubuntu bionic-updates/main amd64 libqt5gui5 amd64 5.9.5+dfsg-0ubuntu2.5 [2,568 kB]
Get:10 http://mirrors.aliyun.com/ubuntu bionic-updates/main amd64 libqt5widgets5 amd64 5.9.5+dfsg-0ubuntu2.5 [2,203 kB]
```

ROS Installation

Step 5. Environment setup

$echo "source /opt/ros/melodic/setup.bash" >> ~/.bashrc
$source ~/.bashrc

Step 6. Install rosinstall

$sudo apt install python-rosdep python-rosinstall python-rosinstall-generator python-wstool build-essential

Step 7. Initialize rosdep

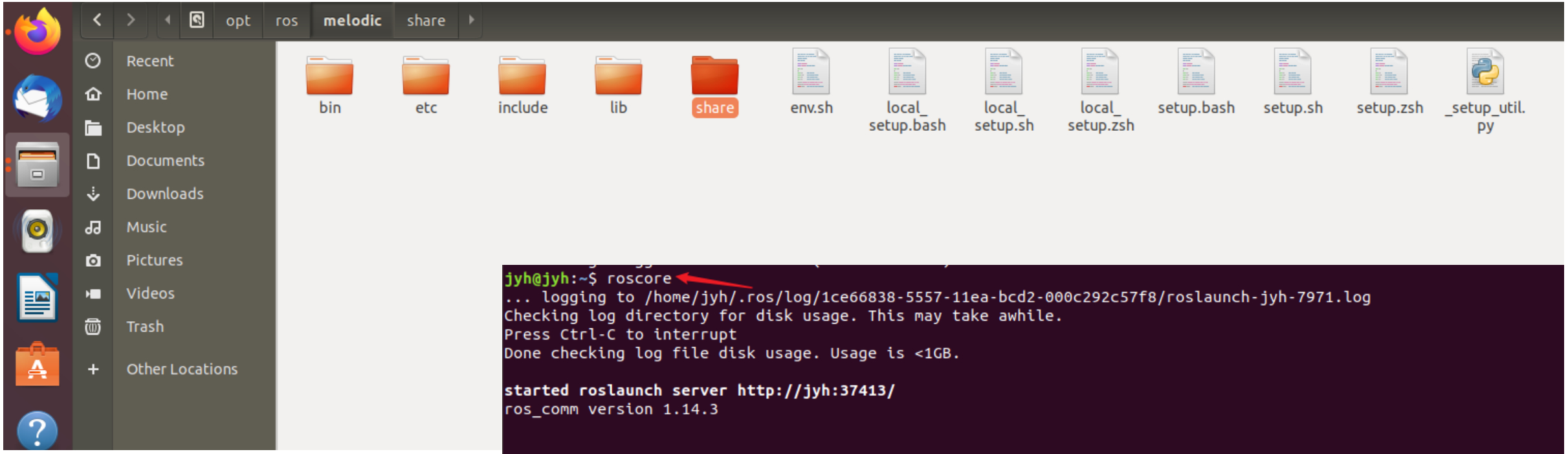
$sudo rosdep init
$rosdep update

```
jyh@jyh:~$ sudo rosdep init
Wrote /etc/ros/rosdep/sources.list.d/20-default.list
Recommended: please run

        rosdep update
```

```
jyh@jyh:~$ rosdep update
reading in sources list data from /etc/ros/rosdep/sources.list.d
Hit https://raw.githubusercontent.com/ros/rosdistro/master/rosdep/osx-homebrew.yaml
Hit https://raw.githubusercontent.com/ros/rosdistro/master/rosdep/base.yaml
Hit https://raw.githubusercontent.com/ros/rosdistro/master/rosdep/python.yaml
Hit https://raw.githubusercontent.com/ros/rosdistro/master/rosdep/ruby.yaml
Hit https://raw.githubusercontent.com/ros/rosdistro/master/releases/fuerte.yaml
Query rosdistro index https://raw.githubusercontent.com/ros/rosdistro/master/index-v4.yaml
Skip end-of-life distro "ardent"
Skip end-of-life distro "bouncy"
Skip end-of-life distro "crystal"
Add distro "dashing"
Add distro "eloquent"
Add distro "foxy"
Skip end-of-life distro "groovy"
Skip end-of-life distro "hydro"
Skip end-of-life distro "indigo"
Skip end-of-life distro "jade"
Add distro "kinetic"
Skip end-of-life distro "lunar"
Add distro "melodic"
Add distro "noetic"
```

```
jyh@jyh:~$ echo "source /opt/ros/melodic/setup.bash" >> ~/.bashrc
jyh@jyh:~$ source ~/.bashrc
jyh@jyh:~$ sudo apt install python-rosinstall python-rosinstall-generator python-wstool build-essential
Reading package lists... Done
Building dependency tree
Reading state information... Done
build-essential is already the newest version (12.4ubuntu1).
build-essential set to manually installed.
The following packages were automatically installed and are no longer required:
  efibootmgr gir1.2-geocodeglib-1.0 libfwup1 ubuntu-web-launchers
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  bzr git git-man liberror-perl libserf-1-1 libsvn1 mercurial mercurial-common python-bzrlib python-conf
  python-keyrings.alt python-launchpadlib python-lazr.restfulclient python-lazr.uri python-oauth python-
Suggested packages:
  bzr-doc bzrtools python-bzrlib.tests git-daemon-run | git-daemon-sysvinit git-doc git-el git-email git
  | meld | tkcvs | mgdiff qct python-mysqldb python-bzrlib-dbg python-kerberos python-pycurl python-conf
  gir1.2-gnomekeyring-1.0 python-fs python-gdata python-keyczar python-testresources python-secretstorag
The following NEW packages will be installed:
  bzr git git-man liberror-perl libserf-1-1 libsvn1 mercurial mercurial-common python-bzrlib python-conf
  python-keyrings.alt python-launchpadlib python-lazr.restfulclient python-lazr.uri python-oauth python-
  python-simplejson python-vcstools python-wadllib python-wstool subversion
0 upgraded, 27 newly installed, 0 to remove and 7 not upgraded.
Need to get 11.4 MB of archives.
After this operation, 67.8 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://mirrors.aliyun.com/ubuntu bionic/main amd64 python-configobj all 5.0.6-2 [34.2 kB]
Get:2 http://mirrors.aliyun.com/ubuntu bionic/main amd64 python-bzrlib amd64 2.7.0+bzr6622-10 [1,456 kB]
Get:3 http://mirrors.aliyun.com/ubuntu bionic/main amd64 bzr all 2.7.0+bzr6622-10 [37.4 kB]
Get:4 http://mirrors.aliyun.com/ubuntu bionic/main amd64 liberror-perl all 0.17025-1 [22.8 kB]
Get:5 http://mirrors.aliyun.com/ubuntu bionic-updates/main amd64 git-man all 1:2.17.1-1ubuntu0.5 [803 kB
Get:6 http://mirrors.aliyun.com/ubuntu bionic-updates/main amd64 git amd64 1:2.17.1-1ubuntu0.5 [3,912 kB
Get:7 http://mirrors.aliyun.com/ubuntu bionic/universe amd64 libserf-1-1 amd64 1.3.9-6 [44.4 kB]
```

ROS Verification




```
jyh@jyh:~$ roscore
... logging to /home/jyh/.ros/log/1ce66838-5557-11ea-bcd2-000c292c57f8/roslaunch-jyh-7971.log
Checking log directory for disk usage. This may take awhile.
Press Ctrl-C to interrupt
Done checking log file disk usage. Usage is <1GB.

started roslaunch server http://jyh:37413/
ros_comm version 1.14.3


SUMMARY
========

PARAMETERS
 * /rosdistro: melodic
 * /rosversion: 1.14.3

NODES

auto-starting new master
process[master]: started with pid [7981]
ROS_MASTER_URI=http://jyh:11311/

setting /run_id to 1ce66838-5557-11ea-bcd2-000c292c57f8
process[rosout-1]: started with pid [7993]
started core service [/rosout]
```

- The default path for the ROS installation directory: ***/opt/ros***
- **Run roscore**

roscore : master (provides name service for ROS) + **rosout** (stdout/stderr) + **parameter server**

ROS Verification

1.Start the ROS master

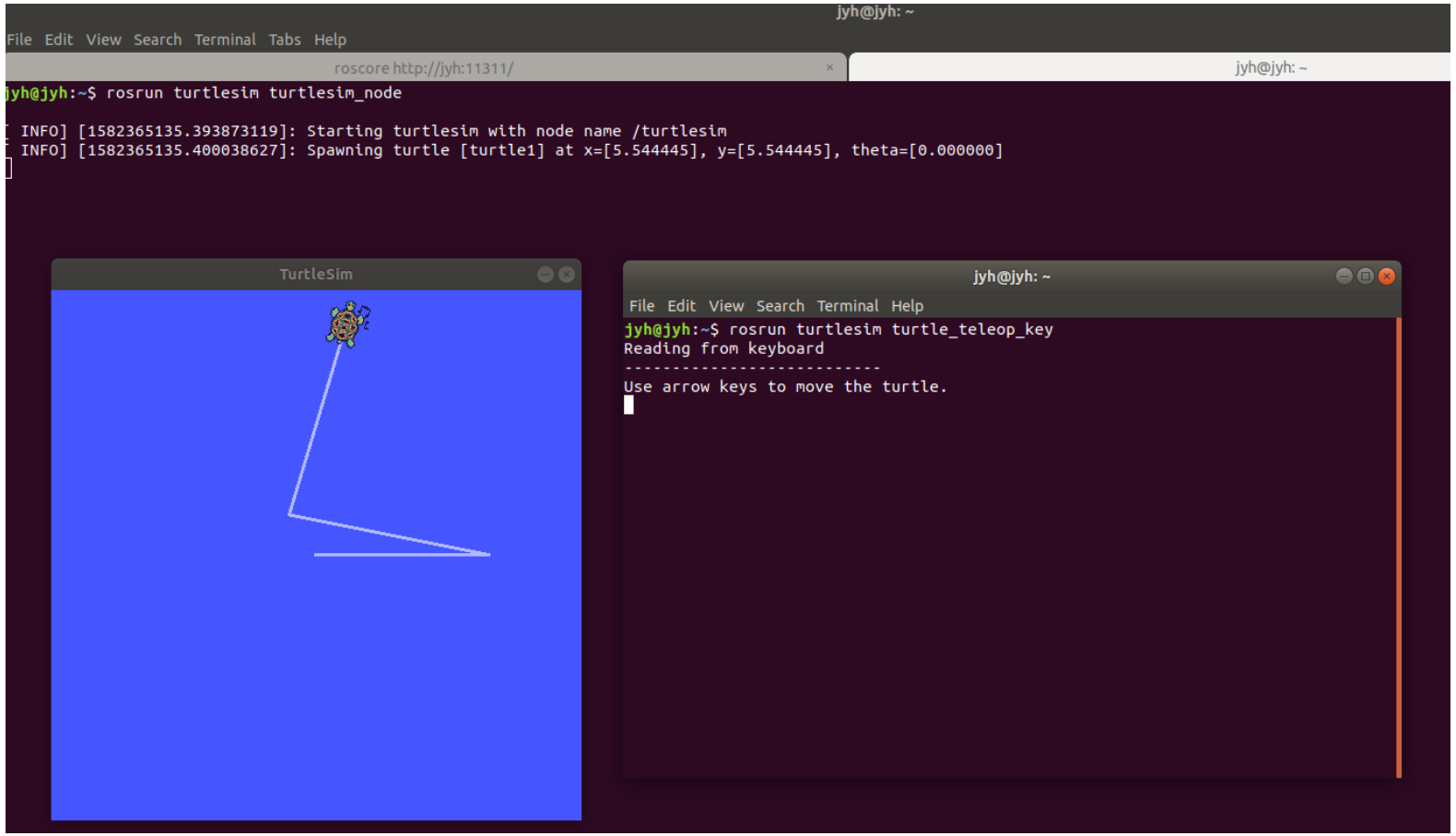
```
$roscore
```

2. Running turtle simulator

```
$rosrun turtlesim turtlesim_node
```

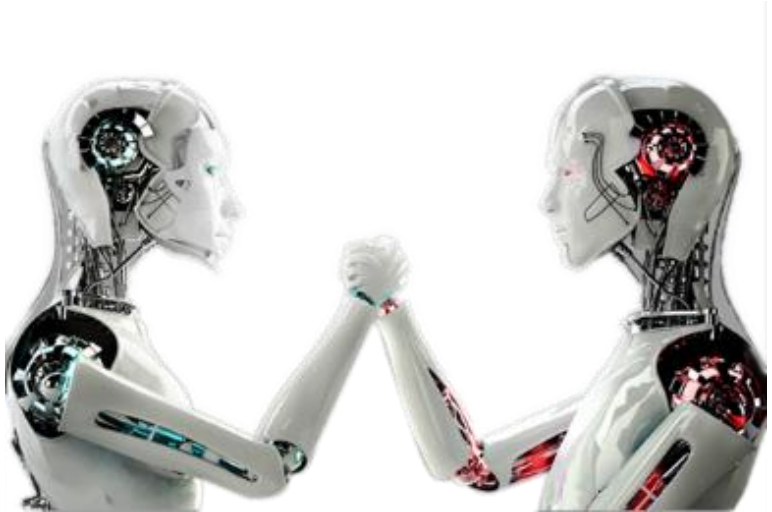
3.Runing control node

```
$rosrun turtlesim turtle_teleop_key
```

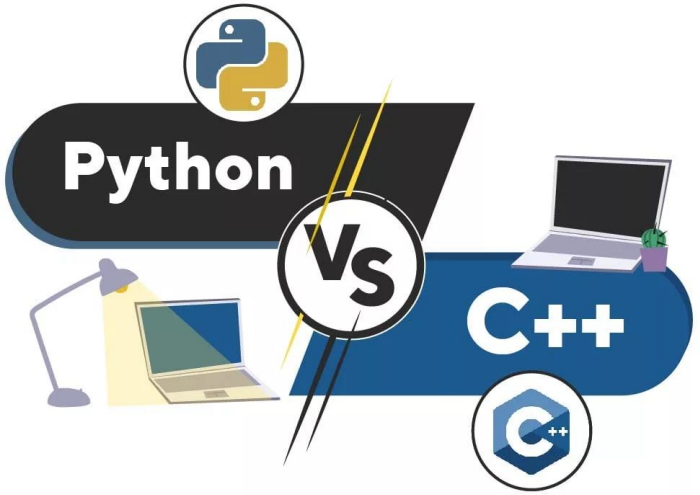


PART FIVE

The basics of C++ & python



The basics of C++ & python



① Ease of learning
② Runtime efficiency
③ Development time

ROS

GETTING STARTED WITH ROS:

C++ or Python?

- Choose one you do best, but the other one must be familiar and able to read the code.
- If you're not familiar with either, here, I recommend you start with python to maximize your ROS learning speed

The basics of C++ & python

Installation

C++:

sudo apt-get install build-essential (*This command installs a bunch of packages, including GCC, g ++, and make*).

sudo apt-get install g++

Python:

sudo apt-get install python

Verification

C++: g++ --version

Python: python --verision

IDE

C++: **Clion or** vscode

Python: **PyCharm or vscode**

The basics of C++ & python

1. if Statement

```cpp
#include <iostream>

using namespace std;

int main( )
{
    // local variable declaration:
    int a = 100;

    // check the boolean condition
    if( a < 20 ) {
        // if condition is true then print the following
        cout << "a is less than 20" << endl;
    } else {
        // if condition is false then print the following
        cout << "a is not less than 20" << endl;
    }
    cout << "value of a is : " << a << endl;
}
```

C++

```python
#!/usr/bin/python
# -*- coding: utf-8 -*-
a=100
if(a<20):
    print("a is less than 20")
else:
    print("a is not less than 20")
print "value of a is :", a
```

python

The basics of C++ & python

2. for Statement

```cpp
#include <iostream>

using namespace std;

int main()
{
    int a=0;
    for(a; a<10; a++)
    {
        cout << "a = " << a << endl;
    }

    return 0;
}
```

C++

```python
#!/usr/bin/python
# -*- coding: utf-8 -*-
for a in range(10):
    if a < 10:
        print 'a = ', a
    else:
        break
```

python

The basics of C++ & python

3. while Statement

```cpp
#include <iostream>
using namespace std;

int main()
{
    int a = 0;

    while(a < 10)
    {
        cout << "a = "  << a << endl;
        a++;
    }

    return 0;
}
```

C++

```python
#!/usr/bin/python
# -*- coding: utf-8 -*-
a = 0
while a < 10:
    print 'a = ', a
    a += 1
```

python

The basics of C++ & python

4.Class

```cpp
#include <iostream>
#include <string>
using namespace std;
class Student
{
public:
    static int stuCount;
    string name;
    int id;
    Student(string name,int id){
        this->name = name;
        this->id = id;
    }
    void displayCount()
    {
        cout <<"Total Student" << Student::stuCount <<endl;
    }
    void displayStudent()
    {

        cout << "Name : "<< name <<", ID: "<<id<< endl;
    }
};
int Student::stuCount = 0;
int main()
{
    Student stu1("Jimmy", 111);
    stu1.displayStudent();
    stu1.displayCount();

    Student stu2("Eddy", 222);
    stu2.displayStudent();
    stu2.displayCount();
    return 0;
}
```

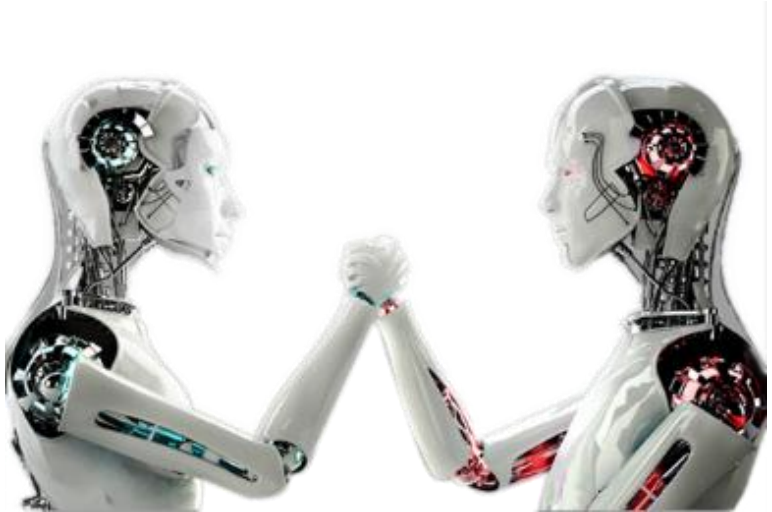
C++

```python
#!/usr/bin/python
# -*- coding: utf-8 -*-
class Student:
    stuCount = 0

    def __init__(self, name, id):
        self.name = name
        self.id = id
        Student.stuCount += 1

    def displayCount(self):
        print "Total Student %d" % Student.stuCount

    def displayStudent(self):
        print "Name : ", self.name, ", ID: ", self.id

def main():
    stu1 = Student("Jimmy", 111)
    stu1.displayStudent()
    stu1.displayCount()

    stu2 = Student("Eddy", 222)
    stu2.displayStudent()
    stu2.displayCount()

if __name__ == "__main__":
    main()
```

python

PART FOUR

Lab Task



Lab Task

1. Install ubuntu on your computer.
2. Practice Linux commands.
3. Install ROS (http://wiki.ros.org/melodic/Installation/Ubuntu)
4. ROS Tutorials (Beginner Level & Intermediate Level) (http://wiki.ros.org/ROS/Tutorials)

Thanks